Final Implementation Report
Team F - AR Aircraft

The minimum viable product was achieved before schedule, which included all critical functions of the project. This means the rendering of aircraft represented by the data received from the Open Sky Network API, including receiving, parsing, updating the data. These aircraft could only be sampled at a minimum of 10 seconds so interpolation was necessary to depict an actual flying plane within X-Plane. These tasks only needed to be integrated with one another once achieved. Objectives which were achieved on time include the outputting of the UDP Stream. The 2D Map and all of its components were implemented on time. The creation and capturing of the Offline Database were also completed on time. Of all the tasks, the only one which explicitly fell behind schedule was the integration of the offline database into the rest of the code. This was as it was a major change to the state of the program and needed extra attention to ensure that it was implemented correctly such that the offline functionality did not kick in when there was only a lull in retrieving data. The only task which was only partially achieved behind schedule was improving the interpolation of the aircraft, which was sorely needed due to the frequent stuttering the aircraft objects would experience when the API data receiving lagged due to the APIs limitations. While there were better ways to implement interpolation, our team opted for simply increasing the offset time in which the program waited before rendering the aircraft within X-Plane, allowing it to better smooth out the rendering when the data was not received consistently. It was possible to implement solutions such as Bezier curves, however for the scope of our project, it was decided that focusing on other aspects of the program was more crucial. There is only one item in the Gantt chart which was canceled, the ICAO lookup table. This component was meant to allow the X-Plane plugin to render the real world aircraft model associated with the ICAO number. It was scrapped due time constraints.